

Top 15 Python Interview Questions

QUICK-REFERENCE GUIDE FOR TECHNICAL INTERVIEWS

1. Difference between a List and a Tuple?

- **List:** Mutable (can be modified), slower, uses more memory, syntax: `[]`.
- **Tuple:** Immutable (cannot be modified), faster, highly memory-efficient, syntax: `()`.

2. How is Memory Management handled in Python?

Managed automatically via **Reference Counting** (tracking references to an object; freeing it when count reaches zero) and a cyclical **Garbage Collector** (destroying objects trapped in circular references).

3. Explain the difference between 'is' and '=='?

- `==` compares the actual **values** of two data objects.
- `is` compares the **memory addresses (identity)** to see if they point to the exact same object.

4. What is a Decorator?

A function that accepts another function as an argument to dynamically extend or modify its behavior without permanently altering its structural source code.

5. What are Python Generators and the 'yield' keyword?

Generators are special iterator functions that use `yield` instead of `return`. They produce elements lazily one at a time, making them highly memory-efficient for large datasets.

6. Difference between Deep Copy and Shallow Copy?

- **Shallow Copy:** Copies the outer object but inserts pointers/references to original inner nested elements.
- **Deep Copy:** Recursively creates completely independent duplicates of both the parent object and all nested elements.

7. What does the '__init__' method do?

It acts as the class constructor. Python triggers it automatically when creating a new instance object to initialize object-specific properties and starting states.

8. Explain '*args' and '**kwargs'.

- `*args` : Receives an arbitrary sequence of optional positional arguments grouped together into a **tuple**.
- `**kwargs` : Receives an arbitrary sequence of optional named arguments grouped together into a **dictionary**.

9. What is a Lambda function?

An anonymous, single-expression function defined inline via the `lambda` keyword. Perfect for short, quick functional operations like sorting rules or filtering arrays.

10. How does Python handle Exception Handling?

Using structural blocks: `try` (runs risky code), `except` (catches/handles exceptions), `else` (runs if zero exceptions occur), and `finally` (always executes for final cleanup tasks).

11. What is PEP 8?

The standard official style guide for writing Python code. It specifies structural guidelines (like using 4 spaces per indent level and `snake_case` names) to maximize codebase readability.

12. Explain the LEGB rule for scoping.

The order Python checks variables: **L**ocal (inside current function) → **E**nclosing (nested functions) → **G**lobal (top level file layer) → **B**uilt-in (native internal Python functions).

13. Difference between 'break', 'continue', and 'pass'?

- `break` : Immediately halts and exits the entire loop.
- `continue` : Skips the rest of the current iteration loop and jumps to the next cycle.
- `pass` : A blank structural placeholder statement that executes nothing.

14. What are List Comprehensions?

A concise, shorthand syntax to instantiate new list sequences out of loops or iterables (e.g., `[x for x in data]`). They offer faster performance and cleaner code.

15. Why is Python "Dynamically Typed"?

Variable types are verified and bound at runtime rather than during compilation. Variable assignments do not explicitly lock a type and can change freely during script execution.